

CMS Hospital Quality Analysis - Findings Summary

I. Problem Statement

- A. This project analyzes publicly available CMS Medicare hospital data to identify regional disparities in hospital quality ratings and patient survey coverage across the United States. Using the Hospital General Information and HCAHPS (Hospital Consumer Assessment of Healthcare Providers and Systems) datasets, the analysis examines how overall ratings vary by region, ownership type, and state, and where underperforming quality measures are most significant. Outputs include a Tableau dashboard and a one-page findings summary.

II. Methodology

- A. CMS public data
 1. Datasets were accessed via the CMS government website at the following links:
 - a) <https://data.cms.gov/provider-data/dataset/xubh-q36u#data-table>
 - b) <https://data.cms.gov/provider-data/dataset/xubh-q36u#data-table>
- B. Data Cleaning
 1. Datasets were cleaned using Microsoft Excel and SQL in BigQuery. The detailed change log is linked below:
 - a) <https://docs.google.com/document/d/1hnPrXkcfColNM2nvYu2o4iIU1iVcXYE1HT8L8AvmjII/edit?usp=sharing>
- C. Analysis
 1. Analysis was performed using SQL in BigQuery. SQL Queries can be found at the link below:
 - a) <https://github.com/nedouard03/CMS-Hospital-Quality-Analysis>

III. Key Insights

- A. The Midwest region outperforms all other regions in both overall quality rating (3.28) and HCAHPS patient survey rating (3.30). The South is the lowest performing region, lagging behind the Midwest by 0.31 points with an overall quality rating of 2.97. US Territories were excluded from this analysis due to a lack of adequate sample population (only 11 hospitals), but they severely underperformed all other regions.
- B. Veteran Health Administration hospitals outperform all other ownership types on both overall quality rating (4.20) and HCAHPS patient survey ratings (3.91), while government-owned and proprietary hospitals show the lowest overall ratings and some of the highest average composite worse counts across mortality, safety, and readmission measures.
- C. Only Mississippi has less than 80% of hospitals with HCAHPS patient survey ratings for facilities with overall quality ratings, with a survey coverage percentage of 73.3%. This indicates patient experience data is available for a significant portion of the hospital that have overall ratings, allowing the completeness of quality measurement in those markets.
- D. 10 states are flagged as High Risk based on a combination of high low-star rating concentration, elevated mortality performance below national average, and elevated safety performance below national average, with Mississippi, New York, Alabama, Kentucky, Georgia, Illinois, Arkansas, New Mexico, West Virginia, and Nevada representing the most acute areas of concern.

IV. Implications

- A. Across various quality measures, the South consistently underperforms and should remain a region of concern. Regarding ownership types, government-owned and proprietary hospitals lag behind other ownership types across overall ratings and quality measures.